

from the contacts, the switch opens. A change in the orientation of the sensor causes the mercury bead to roll in the tube, shorting two electrical contacts and closing the switch. Mercury switch is less popular due to its toxicity.

Ball-in-cage switch. It uses a small metal ball held in a housing unit with two or more contacts. When the housing is tilted, the ball shorts the two electrical contacts, turning on the switch. Multiple contacts can be added to this device so that tilt direction can be determined.

We have used a 4-pin RBS04 tilt sensor. It has the capability to detect horizontal tilt and up-side down orientation.

Circuit description

Fig. 1 shows the circuit for car-bonnet light control using tilt sensor. Pin de-

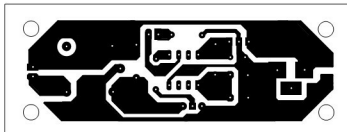


Fig. 3: An actual-size, single-side PCB for car bonnet light control using tilt sensor

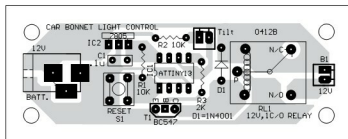


Fig. 4: Component layout for the PCB

tails of ATtiny13, tilt sensor and BC547 are shown in Fig. 2.

ATtiny13 operates at voltages ranging between 1.8V and 5.5V. It

is reset when a low level signal is present on the reset pin (pin 1). Reset pin is made high with resistor R1 for normal operation. Switch S1 provides the manual reset.

Pin 2 of tilt sensor is connected to port pin PB1 of the micro-controller and its pin 4 is connected to R2. Port pin PB0 controls bulb B1 through contacts of relay RL1. When port pin PB1 goes low due to inclination of the tilt sensor, port pin PB0 becomes high. As a result, the transistor conducts and relay RL1 energises to switch on bulb B1. Diode D1 acts as a free-wheeling diode.

When you close the bonnet after

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